Iván Villegas Pérez

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RESEARCH PROFILE

M.Sc. physicist specialized in astroparticle physics and indirect dark matter searches with gamma-ray telescopes. Research experience within the MAGIC Collaboration focused on Bayesian modelling of dark matter density profiles, J-factor computations, and statistical analyses relevant for indirect dark matter searches with Imaging Atmospheric Cherenkov Telescopes. Interested in dark matter phenomenology, γ-ray astrophysics, detector instrumentation, and high-energy particle astrophysics.

RESEARCH INTERESTS

Experimental particle and astroparticle physics, detector instrumentation, signal processing, data analysis for high-energy physics, γ-ray astronomy, computational methods in experimental physics

EDUCATION

University of Oslo (UiO), Oslo, Norway — August 2023 — June 2025
Master of Science in Nuclear and Particle Physics — Cumulative grade: 8.1/10
Thesis Title: Analysis of the dark matter density profile of the globular cluster M15

University of Cantabria (UniCan), Santander, Spain — September 2019 — June 2023
Bachelor of Science: Fundamental Physics — Cumulative grade: 7.93/10
Thesis Title: Simulation, Detection, and Characterization of Exoplanet Transits with the Kepler Space Mission (Simulación, detección y caracterización de tránsitos de exoplanetas con la misión espacial Kepler)

ACADEMIC EXPERIENCE

MAGIC Collaboration — Oslo, Norway
M.Sc. student at UiO — November 2023 — June 2025

- Performed Bayesian Jeans analyses of stellar kinematic data for the globular cluster M15.
- Computed astrophysical J-factors using multiple dark matter halo parametrizations (Einasto, Burkert, Zhao–Hernquist).
- Applied Markov chain Monte Carlo (MCMC) methods and statistical model comparison techniques.
- Contributed to indirect dark matter searches with the MAGIC telescopes.
- Completed a six-week technical shift at the *Observatorio del Roque de los Muchachos* (La Palma, Spain), contributing to telescope operations and data-taking activities.
- Presented results at international collaboration meetings and conferences.
- Co-author and corresponding author of a MAGIC Collaboration manuscript currently in preparation.

Instituto de Física de Cantabria (IFCA) — Santander, Spain
Student Intern — Spring 2023

- Development and characterization of an instrumental calibration source for CMB polarization experiments.
- Work at the interface between instrumentation, calibration, and data analysis.
- Study of systematic effects in detector response.
- Exposure to space-borne and ground-based experimental constraints, directly relevant to the design and calibration of instrumentation for future space-borne missions (e.g. HERD, FOCUS).

PROJECTS

Analysis of the dark matter density profile of the globular cluster M15 — Oslo, Norway
M.Sc. Research project (MAGIC Collaboration) — November 2023 — June 2025

- Computation of astrophysical J-factors for the globular cluster M15 using CLUMPY.
- Implementation and comparison of different dark matter density profiles (Einasto, NFW, cored models).
- Study of tidal effects and their impact on the expected γ-ray flux.
- Interpretation of results in the context of indirect dark matter searches with IACTs.
- Available here: https://hdl.handle.net/10852/120910.

Computing the CMB power spectrum — Oslo, Norway
Course project (Cosmology II, MSc at UiO) — Spring 2024

- Computation of the CMB temperature angular power spectrum using publicly available cosmological data.

- Numerical implementation of spherical harmonic decomposition and power spectrum estimation.
- Comparison of theoretical predictions with observational results.
- Available here: https://github.com/ivanvillegas7/CMB_power_spectrum.

Simulation, Detection, and Characterization of Exoplanet Transits with the Kepler Space Mission Santander, Spain
Bachelor's Thesis (Trabajo de Fin de Grado) September 2022 — June 2023

- Simulation and detection of exoplanet transit signals using photometric data from the *Kepler* space mission.
- Implementation of transit detection algorithms and characterization of planetary parameters.
- Analysis of noise sources and systematic effects in space-based photometric observations.
- Available here: https://hdl.handle.net/10902/30224.

A review of "Calibration Satellite for Ultrasensitive Cosmic Microwave Background Polarization ground-based Experiments" Master's Thesis Santander, Spain
Undergraduate Research Project (IFCA-University of Cantabria) Spring 2023

- Design and characterization of an on-board calibration source for CMB polarization experiments.
- Study of instrumental systematics affecting ground-based CMB telescopes.
- Contribution to feasibility studies for a low Earth orbit (LEO) calibration mission.
- Available here: https://github.com/ivanvillegas7/CalSat_for_Ground-based_CMB_experiments.

Application of Schrödinger's equation in a box using a single, double and triple slit configuration Oslo, Norway
Course project (Computational Physics, BSc at UniCan/ERASMUS at UiO) Fall 2021

- Developed Python-based numerical implementations for solving the time-independent Schrödinger equation.
- Applied numerical methods to study quantum-mechanical systems and eigenvalue problems.
- Implemented scientific visualization tools for wave-functions and energy spectra.
- Available here: https://github.com/ivanvillegas7/Schrodinger_equation.

PUBLICATIONS

Journal Articles

- Saturni, F. G., Maggio, C., Villegas-Pérez, I., D'Amico, G., Doro, M., & Manganaro, M., *Indirect dark matter searches in the globular cluster M15 with the MAGIC telescopes.* Manuscript in preparation for submission **(Corresponding author)**.

CONFERENCE PROCEEDINGS AND CONTRIBUTIONS

- Oral presentation — MAGIC+LST Face-to-Face Science Meeting 2025, Garching, Germany.
- Oral presentation — 28th Nordic Particle Physics Meeting 2025, Svingvoll, Norway.
- Oral presentation — MAGIC General Meeting 2024, Łódź, Poland.

SELECTED COURSES

Master's Courses

- Methods and Instrumentation for Nuclear and Particle Physics
- Particle Physics
- Relativistic Quantum Field Theory
- Cosmology II
- Astroparticle Physics
- Bayesian Cosmological Data Analysis

Bachelor's Courses

- Quantum Mechanics
- Astrophysics
- General Relativity
- Advanced Computation
- Astronomy
- Computational Physics
- Introduction to Nuclear and Particle Physics

AWARDS

Young talents Santander, Spain
Internship contract awarded at SODERCAN

2026 **Erasmus+ Scholarship** Oslo, Norway
Exchange program at the University of Oslo 2022–2023

High Academic Distinction in High School Santander, Spain
Finished High School with a 9.27 GPA 2019

6th place at the Physics Olympics in Cantabria Santander, Spain
Ranked 6th in the Regional Physics Olympiad 2019

OTHER EXPERIENCES

Project Technician in the Business Innovation Area Cantabria, Spain
SODERCAN 2026 — Present

- Support for R&D and innovation projects involving companies and public institutions; technical and administrative project documentation.

Volunteer Dover, England (U. K.)
Pegasus Playscheme 2016

- Adapted communication and activity plans to individual needs, developing strong skills in patience, empathy, and problem-solving under pressure.

SKILLS

- Experimental data analysis.
- Signal processing (time-series, noise, systematics).
- Statistical inference & Bayesian methods.
- Familiarity with large experimental software frameworks.
- Willingness and ability to learn electronics / photonics / ML techniques.

- **Programming & Scientific Computing:** Python, C++, MATLAB, Linux/Unix environments, Git/GitHub, LaTeX.
- **Scientific Tools:** `ROOT, CLUMPY, NumPy, SciPy, Matplotlib, pandas`.
- **Data Analysis:** Statistical modelling, Monte Carlo simulations.
- **Languages:** Spanish (native), English (C1), German (B1) and Norwegian (basic).

REFERENCES

Dr. Giacomo D'Amico
Postdoc researcher, Gamma-ray Group, Institut de Física d'Altes Energies (IFAE), Barcelona, Spain
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